# Real-Time Object Detection and Tracking: Challenges, Innovations and Future Directions

Ahmed Gouda Ismail[1], Mohamed S. Mohamed[1], Tarek Ahmed Mahmoud[2]
[1]Computer Engineering and Artificial Intelligence Department, MTC
[2]Computer Engineering Department, EUI
a7medgouda1@gmail.com, mohamedms@mtc.edu.eg, tarek.mohamed@eui.edu.eg

***Abstract*—real-time Object tracking is generally considered a fundamental, core task in practical scenarios such as healthcare, self-driving cars, monitoring, and robots. This review surveys a systematic taxonomy, a new quantitative comparison and uncovers open scientific questions in view of real-time deployment. It reveals the transition from classical techniques to deep learning and transformer-based approaches, discussing important technical features and solution paradigms as well as presenting a structured and deployment-oriented comparison of the state-of-the-art object detection models and tracking algorithms. In addition, the famous datasets and evaluation metrics are thoroughly discussed giving a wide-angle view on the current developments and pointing out the future research areas.**

***Index Terms*—Object Detection, Object Tracking, Real-Time Systems, Deep Learning, Transformers.**

## I. Introduction

Object detection and tracking have evolved through three major technological phases [1]. In the first phase (before 2012), traditional methods relied on hand-engineered features like Histogram of Oriented Gradients (HOG) and Scale-Invariant Feature Transform (SIFT) [2]. This phase was followed by the second phase (deep learning) that began in 2012 with the introduction of Convolutional Neural Networks (CNN) [3]. During this era, models like R-CNN and YOLO expanded the speed-accuracy frontiers [4], [5]. The third phase, beginning around 2020, introduces transformer-based vision models [6]. This has led to DETR, TransTrack [7], and other end-to-end models which consolidate the whole detection and tracking pipeline in a single step [8], [9]. Unlike earlier surveys, this work provides a unified, deployment-oriented perspective [1].

The main contributions of this work can be summarized as follows:

- A structured taxonomy covering traditional, deep learning-based, and transformer-driven detection and tracking approaches.
- An up-to-date quantitative comparison of state-of-the-art detectors and trackers under real-time constraints.
- A deployment-focused discussion that jointly analyzes accuracy, latency, model size, and hardware suitability.
- Identification of open challenges and future research directions for real-time and edge-based tracking systems.

## II. Real-Time Object Detection and Tracking Applications

Real-time object detection and tracking are essential for applications requiring accurate environmental perception [1], [10]. Key domains include surveillance and security systems for automatic abnormal behavior detection. You Only Look Once (YOLO)-based models are widely used for real-time object and human detection in live video streams [11]. In addition, AI-based surveillance systems employ deep learning to recognize and classify objects such as weapons or suspicious packages [3]. Autonomous vehicles and Advanced Driver Assistance Systems (ADAS) rely on real-time detection to identify lane markings, traffic signs, pedestrians, and obstacles [12]. Unmanned Aerial Vehicles (UAVs) apply these techniques in defense, environmental monitoring, and search operations, commonly using Oriented FAST and Rotated BRIEF (ORB) features and YOLO detectors [10]. Industrial robotics applications include pick-and-place operations, assembly, and quality control, where feature-based tracking supports navigation and Simultaneous Localization and Mapping (SLAM) [1].

## III. Object Detection and Tracking Technical Issues

Real-time object detection and tracking faces some technical challenges that affect deployment performance in unconstrained real-world environments [1].

### A. Detection and Tracking Challenges

In crowded scenes, distinguishing individual objects becomes difficult due to occlusions and identity switches. Fast motion, deformation, and long-term re-identification require robust appearance modeling [2]. Multi-object tracking faces three main challenges: identity switches, data association across frames, and crowded scene handling with frequent occlusions [1], [13].

### B. Real-time Constraints

Real-time performance in computer vision is mainly limited by three factors [10]. The accuracy-speed trade-off suggests that the more accurate a model is, the more computation it needs, and thus the longer the inference time [11]. Computational cost depends on the hardware, ranging from high-end GPUs for cloud-based applications to low-power edge AI devices for on-device processing [10]. The deployment restrictions point to different environmental factors like network bandwidth, power

supply, and remote or offline processing [1]. The resolution of these technical issues will pave the way for a significant amount of innovation in model architectures, training methodologies, and hardware optimization, and will often require making the trade-offs between accuracy, robustness, and computational efficiency [14].

### C. Formal Problem Definition

Let a video sequence be represented as $\{I_t\}_{t=1}^{T}$, where $I_t$ denotes the frame at time $t$. Object detection aims to estimate a set of bounding boxes $\mathcal{D}_t = \{(b_i^t, c_i^t, s_i^t)\}$, where $b_i^t$ is the bounding box, $c_i^t$ the class label, and $s_i^t$ the confidence score [4].

Multi-object tracking seeks to associate detections across time to form trajectories $\mathcal{T} = \{\tau_j\}$, where each trajectory is defined as $\tau_j = \{b_j^t\}_{t=t_s}^{t_e}$, where $t_s$ and $t_e$ denote the start and end time indices of the trajectory, respectively [13]. The core challenge lies in solving the data association problem:

$$\arg\min_{\pi} \sum_{t} \mathcal{C}(\mathcal{D}_t, \mathcal{T}_{t-1}),$$

where $\mathcal{C}$ represents a cost function based on motion, appearance, or learned embeddings [15]. This formulation highlights the intrinsic dependency between detection accuracy and association reliability in real-time tracking systems [1].

## IV. Overview of Solution Approaches

### A. Tracking-by-Detection Paradigm

Tracking-by-detection (TBD) is the dominant paradigm in multi-object tracking, where object detectors are applied independently per frame and detections are associated over time using motion and appearance cues [15], [16]. This paradigm benefits from advances in standalone detectors while remaining modular and flexible [1].

### B. Joint Detection and Tracking Approaches

Joint detection and tracking (JDT) approaches aim to overcome some limitations of the TBD paradigm by integrating the detection and tracking processes into a single, end-to-end trainable framework [17]. Detection and tracking often have very similar characteristics among the methods described, leading to more consistent and quicker processing. CenterTrack and FairMOT are just two of the examples that directly estimate the identities and motions of objects, thereby cutting down the dependence on the separate steps for association [18].

### C. Object Tracking using Deep Learning

The latest detectors comprising YOLO and R-CNN groups are constructed on top of the CNN architecture [3]. They develop layered representations starting from raw image data, hence are able to detect intricate patterns and features that are essential for both object localization and classification [4], [19]. Graph Neural Networks (GNNs) are an architecture that is still in the making in object tracking; however, they are very helpful in the process of modeling complex relationships among multiple objects in a scene [1]. They are able to portray objects as nodes and their interactions as edges which further supports data association and re-identification to be more sophisticated especially in crowded and dynamic environments [1].

### D. Transformer-based Tracking

Transformer-based methods treat tracking as set prediction, directly predicting object states [6]. TrackFormer and TransTrack achieve strong performance in challenging scenarios [7], [8]. End-to-end transformer models eliminate the need for hand-crafted components such as Kalman filters or IoU-based matching, offering a fully differentiable training pipeline [9].

## V. Evaluation Metrics

### A. Object Detection Metrics

**Frames Per Second (FPS):** FPS is a figure that reflects the speed with which the object detection model performs inference and it is defined as the number of frames that are processed in a second. It is a very crucial metric in real-time applications where the higher the FPS, the faster the processing and the lower the latency [10].

**Average Precision (AP):** Precision measures the fraction of correct detections among all detections, while recall measures the fraction of ground-truth objects that were successfully detected. AP summarizes the precision-recall curve as a single value by computing the weighted mean of precisions at different recall thresholds [20]. For a given class and IoU threshold, AP is calculated by integrating the area under the precision-recall curve [20].

**Mean Average Precision (mAP):** mAP is the primary metric for object detection models. It is calculated by averaging the Average Precision (AP) over all object classes and often across different IoU thresholds. A higher mAP indicates better detection accuracy across various object sizes and conditions [10], [20].

### B. Multi-Object Tracking Metrics

**Identity Switches (IDS):** This parameter simply counts the instances wherein a monitored object is assigned a different identity, and thus lower values are better for tracking consistency [1].

**IDF1 Score:** This metric is a harmonic mean of Identity Precision (IDP) and Identity Recall (IDR), which are derived from the correctly identified detections [1].

**Higher Order Tracking Accuracy (HOTA):** HOTA is the latest and most comprehensive metric created to specifically counterbalance the precision in detection, association and localization [13].

**Multiple Object Tracking Accuracy (MOTA):** MOTA is the most common metric used for evaluation which combines the three error sources: FPs, FNs and IDS. MOTA gives one value that mirrors the total tracking performance, with better accuracy shown by higher values [1].

## VI. Object Detection Models Comparison

Object detection architectures are broadly categorized into two paradigms: two-stage and single-stage detectors [4], [21].

### A. Two-Stage Detection Architectures

The Region-based Convolutional Neural Network (R-CNN) family comprises two-stage object detectors known for their high accuracy, particularly in small object detection and complex scenes [4]. While these detectors are generally slower than single-stage methods, their superior accuracy makes them suitable for applications where precision is critical. Faster R-CNN is widely adopted in industries such as healthcare for medical imaging applications [19].

**Faster R-CNN:** Faster R-CNN introduces a Region Proposal Network (RPN) that shares convolutional features with the detection network, enabling efficient and accurate object detection [4]. The RPN generates high-quality region proposals, which are then classified and refined by the detection head. This architecture significantly improves both speed and accuracy compared to earlier R-CNN variants [4].

**Mask R-CNN:** Building on the Faster R-CNN architecture, Mask R-CNN extends the model by adding a parallel branch for predicting segmentation masks alongside bounding boxes and class labels [19]. This enables simultaneous object detection and instance segmentation, finding application in medical image analysis, autonomous driving, and retail analytics [19].

### B. Single-Stage Detection Architectures

**YOLO Family (v3–v12):** The You Only Look Once (YOLO) family of models has been celebrated for their remarkable speed and accuracy, thus they are perfect for real-time applications [11]. YOLO has gone through many changes from the very first release. Each version brought along its own architectural modifications and performance improvements [5], [11], [22]. YOLO models view object detection as a regression problem that involves predicting bounding boxes and class probabilities all at once across a grid [11].

**SSD:** The Single Shot MultiBox Detector (SSD) is a fast, single-stage detector that uses one deep neural network to predict bounding boxes and the class probabilities simultaneously, taking advantage of the feature maps produced at different scales [21]. SSD is frequently used in applications that require high performance, such as video surveillance and mobile phone applications [21].

**RetinaNet:** The introduction of Focal Loss by RetinaNet was to deal with the extremely high class imbalance of foreground and background during the training of dense detectors [23]. Thus, the performance of RetinaNet was equal to the accuracy of two-stage detectors, while it did not lose the speed of one-stage methods. It works particularly well for small object detection in cluttered scenes [23].

**EfficientDet:** The family of EfficientDet consists of object detectors that are both scalable and highly efficient [24]. The introduction of BiFPN (Bi-directional Feature Pyramid Network) made the process of multi-scale feature fusion very efficient. EfficientDet models are capable of reaching high accuracy with dramatically lower parameters and FLOPs than other detectors [24].

**DETR vs. RF-DETR:** DETR (Detection Transformer) introduced a transformer-based paradigm for object detection by eliminating anchor boxes and non-maximum suppression (NMS), enabling a fully end-to-end detection pipeline [6]. Despite its conceptual simplicity, DETR suffers from high computational cost and slow inference, particularly when using large backbones such as ResNet-101 [6]. RF-DETR addresses these limitations by preserving the core DETR formulation while introducing lightweight architectural optimizations [25]. As reflected in Table I, RF-DETR achieves significantly higher detection accuracy with drastically reduced FLOPs, inference latency, and model size, making it well-suited for real-time and practical deployment scenarios [25].

Moreover, we show a detailed comparison of performances from various detection architectures, highlighting the evolution of the YOLO family and other notable detectors. All performance metrics are sourced from official repositories or standardized benchmark evaluations conducted under controlled conditions. It could be noticed that, the two-stage detectors such as Faster R-CNN [4] and Mask R-CNN [19] have comparable accuracy but they are very slow (172–200 ms), making them not suitable for real-time applications. The early single-stage detectors (SSD [21] and YOLOv3 [5]) allow faster inference but exhibit a significant drop in detection accuracy. The recent generations of the YOLO family [11] have a very good accuracy–speed trade-off where the larger variants (YOLO11x, YOLO12x) achieve high mAP at the cost of larger model size and latency, while the lightweight models are fast but with lower accuracy. The performance of EfficientDet [24] is high in terms of detection accuracy but still requires heavy computational power compared to the modern real-time detectors. DETR (R101) [6] shows high latency and parameter count along with limited accuracy gains. RF-DETR-S [25] achieves competitive accuracy (52.9 mAP) with efficient inference (3.5 ms) and a moderate model size (32.1M parameters), demonstrating a favorable balance between accuracy and computational efficiency. Overall, the YOLO family and compact transformer-based models represent the most practical choices for real-time deployment.

## VII. Comparative Analysis of Tracking Algorithms

### A. Traditional and Deep Learning Methods

**SORT (Simple Online and Realtime Tracking):** Being one of the basic methods used in multiple objects tracking, SORT provides an effective baseline that is based on a combination of Kalman filtering for predicting the state and the Hungarian algorithm for the association of detection-to-track [15]. The main benefits of SORT are computational efficiency and simplicity of implementation, which remarkably contribute to the real-time performance of 143.3 FPS. Despite that, the sole dependency on motion cues leads to considerable breakdowns during occlusions and in situations where the background is very dynamic, causing a large number of identity switches (4,852) [15].

**DeepSORT:** This technique particularly focused on the inadequate identity preservation performance of SORT, thereby integrating the deep feature embedding based appearance infor-

TABLE I
OBJECT DETECTION MODELS ON COCO VAL2017. **OFFICIAL SOURCES:** ULTRALYTICS DOCS, ORIGINAL PAPERS FOR CLASSICAL DETECTORS. **HARDWARE:** T4 GPU, TENSORRT FP16, 640×640, BATCH 1.

| Year | Model | mAP | Speed (ms) | Params (M) | Source |
|---|---|---|---|---|---|
| 2015 | Faster R-CNN (R50) | 37.0 | 172 | 41.8 | [4] |
| 2018 | Mask R-CNN (R50) | 37.9 | 200 | 44.2 | [19] |
| 2018 | YOLOv3 | 31.0 | 28.6 | 61.9 | [5] |
| 2020 | YOLOv4 | 43.5 | 15.4 | 64.3 | [22] |
| 2020 | YOLOv5s | 37.4 | 7.1 | 7.2 | [11] |
| 2023 | YOLOv8n | 37.3 | 3.2 | 3.2 | [11] |
| 2023 | YOLOv8s | 44.9 | 12.5 | 11.2 | [11] |
| 2023 | YOLOv8m | 50.2 | 25.9 | 25.9 | [11] |
| 2024 | YOLO11n | 39.5 | 1.5 | 2.6 | [11] |
| 2024 | YOLO11s | 47.0 | 2.5 | 9.4 | [11] |
| 2024 | YOLO11m | 51.5 | 4.7 | 20.1 | [11] |
| 2024 | YOLO11x | 54.7 | 11.3 | 56.9 | [11] |
| 2025 | YOLO12n | 40.6 | 1.64 | 2.6 | [11] |
| 2025 | YOLO12s | 48.0 | 2.61 | 9.3 | [11] |
| 2025 | YOLO12m | 52.5 | 4.86 | 20.2 | [11] |
| 2025 | YOLO12x | 55.2 | 11.79 | 59.1 | [11] |
| 2015 | SSD-512 | 26.6 | 38.2 | 26.3 | [21] |
| 2017 | RetinaNet (R101) | 37.8 | 122 | 56 | [23] |
| 2020 | EfficientDet-D0 | 34.3 | 38.1 | 3.9 | [24] |
| 2020 | DETR (R101) | 42.9 | 145.0 | 60.6 | [6] |
| 2025 | RF-DETR-S | 52.9 | 3.5 | 32.1 | [25] |

mation into the process [16]. The process of association merges two complementary sources: predictions of movement from Kalman filtering and calculating the similarity in appearance by using the cosine distance between embeddings as the measure. This dual-modality technique has remarkably brought down the number of identity switches (from 4,852 to 1,455) [16].

**JDE (Joint Detection and Embedding):** JDE is a novel methodology that combines object detection and appearance embedding in a single network in a unified way [17]. JDE has improved computational efficiency by sharing the feature extraction between detection and re-identification tasks and at the same time has maintained competitive tracking performance. The method produces 64.4% MOTA and 1,544 identity switches [17].

**FairMOT:** FairMOT uses a single network that has common feature extraction for all tasks and separate heads for detection and appearance embedding which are connected in parallel [18]. The main breakthrough is to consider detection and re-identification as tasks of equal significance in the training process. This design of the architecture is very useful especially in the case of crowded situations, thereby reaching 67.5% MOTA with only 953 identity switches [18].

**ByteTrack:** The system of ByteTrack performs a two-part association procedure: at first, it associates the high-confidence detection to the existing tracks, and then it performs second association among the remaining tracks and the low-confidence detections [26]. This tiered technique proves to be very effective in difficult cases of partial occlusions, achieving 80.3% MOTA with an identity switch of 2,196 [26].

**OC-SORT (Observation-Centric SORT):** OC-SORT, specifically designed for occlusion handling, refines the traditional SORT methodology and introduces observation-centric state estimation and consistency in velocity direction [27]. When evaluated using standardized public detections, it achieved 58.2% MOTA with the lowest number of identity switches (784), demonstrating superior resilience to occlusions [27].

**BoT-SORT (Bootstrapping and Temporal Association):** This approach to tracking employs a combination of camera motion compensation (CMC), optimized Kalman filtering, and deep appearance features [28]. BoT-SORT gives an outstanding performance (80.5% MOTA, 80.2% IDF1) with good identity preservation (1,194 switches), being one of the most balanced methods in the traditional category [28].

**Deep OC-SORT:** Deep OC-SORT, an enhancement of OC-SORT, uses deep appearance features to ameliorate re-identification even in occlusion cases [29]. It uses the motion modeling based on observation and the previously learned representation of appearance together, getting the results of 77.6% MOTA, 76.4% IDF1, and 945 identity switches [29].

**StrongSORT:** StrongSORT introduces a variety of elements that bolster DeepSORT, such as exponential moving average (EMA) for appearance features, Non-maximum Suppression with Appearance (NSA), and Enhanced Correlation Coefficient (ECC) for camera motion compensation [30]. This all-encompassing strategy secures 79.6% MOTA and 79.5% IDF1 with 1,024 identity switches [30].

### B. Transformer-Based Methods

**TransTrack:** TransTrack proposes the use of an architecture based on transformers that extracts features from the images sequentially and then predicts the bounding boxes of the objects and assigns identities across the frames directly [31]. The system based on attention is very good at capturing long-range dependencies, but it requires more resources (13.0 FPS) and produces more identity switches (3,663), achieving 75.2% MOTA [31].

**TrackFormer:** In the TrackFormer model, object queries are divided into those for the detection of newly appearing objects and track queries for the continuation of existing trajectories [8]. This dual-query mechanism is especially useful for the maintenance of long-term identity consistency. TrackFormer achieves 74.1% MOTA and 68.0% IDF1 at reduced inference speed (8.0 FPS) [8].

**MOTR (Multiple Object Tracking with Transformers):** MOTR is a complete transformer-based tracking framework that does not require any hand-crafted components [9]. It utilizes track queries that remain across frames and are modified by self-attention and cross-attention mechanisms, achieving 73.4% MOTA and 68.6% IDF1 with 2,439 identity switches [9].

### C. Recent State-of-the-Art Methods

**HybridSORT:** HybridSORT shows a hybrid solution merging the strengths of motion prediction and appearance modeling with better association strategies [32]. HybridSORT achieved 77.8% MOTA and 77.6% IDF1 with 1,169 identity switches at practical inference speed (22.0 FPS) [32].

**SMILEtrack (SiMIlarity LEarning for MOT):** SMILEtrack innovatively resets the similarity learning methodology by introducing a framework that fully integrates detection

TABLE II
PERFORMANCE EVALUATION OF MOT ALGORITHMS ON THE MOT17 BENCHMARK USING IDENTICAL PUBLIC FASTER R-CNN DETECTIONS FOR FAIR COMPARISON.

| Year | Algorithm | MOTA↑ | IDF1↑ | HOTA↑ | FPS | ID Sw.↓ | Frag.↓ | Innovation | Source |
|---|---|---|---|---|---|---|---|---|---|
| *Traditional and Deep Learning Methods* | | | | | | | | | |
| 2016 | SORT | 43.1 | 39.8 | 34.0 | 143.3 | 4,852 | 7,127 | Kalman+Hungarian | [15] |
| 2017 | DeepSORT | 59.8 | 61.4 | – | 40.0 | 1,455 | 2,474 | Deep appearance | [16] |
| 2020 | JDE | 64.4 | 55.8 | – | 30.0 | 1,544 | – | Embedding learning | [17] |
| 2021 | FairMOT | 67.5 | 70.0 | – | 25.9 | 953 | – | Anchor-free JDT | [18] |
| 2022 | ByteTrack | 80.3 | 77.3 | 63.1 | 30.0 | 2,196 | 2,277 | Low-conf assoc. | [26] |
| 2022 | OC-SORT | 58.2 | 65.1 | 52.4 | 28.6 | 784 | 2,006 | Occlusion handling | [27] |
| 2022 | BoT-SORT | 80.5 | 80.2 | 65.0 | 28.0 | 1,194 | 1,298 | CMC+ReID | [28] |
| 2023 | Deep OC-SORT | 77.6 | 76.4 | 62.1 | 20.3 | 945 | 1,619 | Deep occlusion | [29] |
| 2023 | StrongSORT | 79.6 | 79.5 | 64.4 | 19.5 | 1,024 | 1,401 | ECC+NSA+ReID | [30] |
| *Transformer-Based Methods* | | | | | | | | | |
| 2021 | TransTrack | 75.2 | 63.5 | – | 13.0 | 3,663 | – | Query propagation | [31] |
| 2022 | TrackFormer | 74.1 | 68.0 | 58.5 | 8.0 | 2,829 | 4,492 | DETR-based E2E | [8] |
| 2022 | MOTR | 73.4 | 68.6 | 57.8 | 9.5 | 2,439 | 4,092 | Track queries | [9] |
| *Recent State-of-the-Art Methods* | | | | | | | | | |
| 2024 | HybridSORT | 77.8 | 77.6 | 63.2 | 22.0 | 1,169 | 1,580 | Hybrid motion | [32] |
| 2024 | SMILEtrack† | 81.7 | 80.3 | 66.8 | 17.3 | 1,017 | 1,245 | SDE tracking | [33] |
| 2024 | C-BIoU | 80.6 | 79.7 | 65.3 | 25.8 | 1,089 | 1,347 | Complete IoU | [34] |

and association via a single loss function [33]. SMILEtrack maintained cutting-edge performance at 81.7% MOTA, 80.3% IDF1, and 66.8% HOTA, with identity switches remarkably low at 1,017 [33].

**C-BIoU (Complete-Intersection over Union):** C-BIoU suggests an enhanced IoU metric that includes complete geometric relationships between boxes, such as aspect ratio and center point distance [34]. C-BIoU reaches 80.6% MOTA and 79.7% IDF1 at 1,089 identity switches while performing efficiently (25.8 FPS) [34].

The comparison illustrates the tracking paradigms with their respective advantages and disadvantages. The traditional method SORT [15] has the highest speed (143.3 FPS) but also loses the most identities (4,852 switches). Deep learning methods not only increase the accuracy of the tracking but also improve identity consistency, with methods like ByteTrack [26] and BoT-SORT [28] having more than 80% MOTA while being fast enough for practical use. Transformer-based techniques [8], [9] provide theoretical sophistication but at the cost of high computational power. The newest tracking techniques prove that combining cues cleverly and using end-to-end learning can result in state-of-the-art performances, with SMILEtrack [33] being the best at 81.7% MOTA.

### D. Performance Analysis and Key Insights

Table II provides a fair and unified comparison of multi-object tracking algorithms under identical evaluation conditions using public Faster R-CNN detections on the MOT17 benchmark [35]. The results clearly show the evolution of tracking performance over time, highlighting the transition from motion-only methods to appearance-aware and hybrid association strategies [1]. While classical methods such as SORT [15] achieve very high inference speed, they suffer from severe identity instability. Modern tracking-by-detection approaches, including ByteTrack [26] and BoT-SORT [28], offer a strong balance between accuracy, identity preservation, and real-time performance. Transformer-based methods [8], [9] demonstrate conceptual advantages but remain computationally expensive under standardized settings. Recent methods such as SMILEtrack [33] and C-BIoU [34] achieve the best overall performance, confirming that effective integration of appearance modeling, association strategies, and efficiency-aware design is key to state-of-the-art tracking.

## VIII. Datasets for Object Detection and Tracking

### A. Foundational Detection and Tracking Datasets

**COCO (Common Objects in Context):** The dataset gained recognition as a mainstay in the area of object detection and segmentation processing and it consisted of more than 330,000 pictures with very extensive annotations covering 80 object categories [20]. The variety in the sizes of the objects, scenarios, and positions is what makes it a great resource for the training of powerful detection models [20]. Available at: https://cocodataset.org.

**ImageNet VID:** Major video dataset includes more than a million frames that are taken from around 4,000 video clips mostly from YouTube and similar sources. The dataset shows many different real-life conditions with a lot of changes in lighting, camera movement, and the looks of the object. Available at: https://image-net.org/challenges/LSVRC/.

**KITTI:** The KITTI dataset was explicitly meant for autonomous vehicle research, and it gives away thorough information in the form of multi-modal sensor data [12]. Synchronized records of stereo camera, LiDAR point clouds, and GPS/IMU sensors are combined to form the dataset. Available at: https://www.cvlibs.net/datasets/kitti/.

**MOT17 (Multiple Object Tracking Benchmark 2017):** The dataset is made up of 14 video sequences recorded from both stationary and moving cameras [35]. A significant aspect of MOT17 is that it gives access to the public detections from

TABLE III
COMPREHENSIVE OVERVIEW OF OBJECT DETECTION AND TRACKING DATASETS. RECENTLY INTRODUCED UAV-FOCUSED DATASETS (AERIALMIND AND M3OT) ARE INCLUDED BASED ON THEIR OFFICIAL PUBLICATIONS AND RELEASED BENCHMARKS.

| Dataset | Year | Size | Application |
|---|---|---|---|
| COCO [20] | 2014 | 330K images | General detection |
| ImageNet VID | 2015 | 1M+ frames | Video detection |
| KITTI | 2012 | 15K images | Autonomous driving |
| MOT17 [35] | 2017 | 14 sequences | Pedestrian tracking |
| MOT20 [36] | 2020 | 8 sequences | Crowded scenes |
| VisDrone | 2019 | 288 videos, 262K frames | Aerial/UAV tracking |
| AerialMind | 2025 | 93 seqs., 24.6K expr. | Referring MOT (UAV) |
| M3OT | 2025 | Multi-modal | Multi-sensor UAV |

three kinds of detectors (Faster R-CNN, DPM, SDP), which allows the tracking algorithms being compared to be evaluated quite independently of the detection quality [35]. Available at: https://motchallenge.net/data/MOT17/.

**MOT20 (Multiple Object Tracking Benchmark 2020):** The dataset consists of 8 video sequences along with thorough ground truth annotations [36]. Compared to MOT17, the identity preservation and occlusion handling challenges are escalated in MOT20 [36]. Available at: https://motchallenge.net/data/MOT20/.

### B. Specialized and Emerging Datasets

**VisDrone:** The aerial perspective dataset presents the unique challenges of UAV-based surveillance and tracking [10]. With a total of 288 videos and more than 262,000 annotated frames, VisDrone presents a wide range of situations recorded from drones at different altitudes and angles. Available at: https://github.com/VisDrone/VisDrone-Dataset.

**AerialMind:** Introduced in 2025, AerialMind advanced UAV-based tracking research by incorporating natural language-guided object tracking with referring expressions. The dataset contains 93 video sequences with 24.6K referring expressions and over 293K annotated object instances, dedicated to referring multi-object tracking tasks. Available at: https://github.com/shawnliang420/AerialMind.

**M3OT:** M3OT, also available in 2025, has been designed for multi-modal tracking from UAV perspectives and combines RGB imagery with the outputs of other sensors. Available at: https://github.com/M3OT/M3OT.

The characteristics of these datasets as well as their major application areas are presented in Table III. The passage from universal datasets such as COCO [20] to specialized benchmarks like MOT17 [35] and emerging multi-modal datasets showcases the progress of the discipline to deal with more challenging and varied tracking cases [1]. Notably, the recent introduction of UAV-focused datasets (AerialMind, M3OT) in 2025 reflects the growing importance of aerial tracking and language-guided systems as frontier research areas.

## IX. Limitations and Failure Cases

Despite significant progress, real-time object detection and tracking systems remain vulnerable to failure under severe occlusion, extreme illumination changes, and long-term identity re-appearance [1], [2]. Crowded environments often induce identity switches, while low-frame-rate or compressed video streams degrade motion prediction reliability [1]. Moreover, transformer-based methods [8], [9], although powerful, suffer from high computational cost, limiting their applicability in edge deployments [14].

## X. Conclusion and Future Directions

This in-depth survey presented the real-time object detection and tracking status, covering classical techniques through deep learning-based and transformer-based approaches [1]. The family of YOLO models [11] has been leading the way for real-time applications, while R-CNN variants [4], [19] continue to be the best choice in accuracy-critical areas. The object tracking field continues to be heavily influenced by the TBD paradigm [15], [16], with methods like ByteTrack [26] leveraging low-confidence detections for improved robustness. Tracking techniques based on transformers [8], [9], [31], although more demanding in terms of computation, suggest a gradual shift towards integrated detection and tracking systems. Recent state-of-the-art methods such as SMILEtrack [33] confirm that end-to-end similarity learning is a key direction for future progress.

The future research paths include using vision transformers and foundation models, which enable zero-shot and language-guided tracking [14]. Another hopeful direction is 3D multi-modal tracking through the integration of LiDAR, radar, and RGB data to create robust perception under any weather condition [12]. Additionally, the advanced fusion of different sensors comprising visual, thermal, radar, and audio signals will be one of the crucial factors making adverse conditions less of an issue.

## References


[1] S. Li, H. Ren, X. Xie, and Y. Cao, "A review of multi-object tracking in recent times," *IET Computer Vision*, vol. 19, no. 1, 2025.

[2] A. Alashbi, A. H. H. M. Mohamed, A. A. El-Saleh, I. Shayea, M. S. Sunar, Z. R. Alqahtani, F. Saeed, and B. Saoud, "Human face localization and detection in highly occluded unconstrained environments," *Engineering Science and Technology, an International Journal*, vol. 61, p. 101893, 2024.

[3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778.

[4] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, 2017.

[5] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," arXiv:1804.02767, 2018.

[6] N. Carion, F. Massa, G. Synnaert, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. ECCV*, 2020, pp. 213–229.

[7] P. Sun, J. Cao, Y. Jiang, R. Zhang, E. Xie, Z. Yuan, C. Wang, and P. Luo, "TransTrack: Multiple object tracking with transformer," arXiv:2012.15460, 2020.

[8] T. Meinhardt, A. Kirillov, L. Leal-Taixé, and C. Feichtenhofer, "TrackFormer: Multi-object tracking with transformers," in *Proc. IEEE/CVF CVPR*, 2022, pp. 8844–8854.

[9] F. Zeng, B. Dong, Y. Zhang, T. Wang, X. Zhang, and Y. Wei, "MOTR: End-to-end multiple-object tracking with transformer," in *Proc. ECCV*, 2022, pp. 659–675.

[10] M. Bakirci, "Performance evaluation of low-power and lightweight object detectors for real-time monitoring in resource-constrained drone systems," *Engineering Applications of Artificial Intelligence*, 2025.

[11] Ultralytics, "YOLO Documentation," 2020–2025. [Online]. Available: https://docs.ultralytics.com

[12] Y. Chae, H. Kim, and K.-J. Yoon, "Towards robust 3D object detection with LiDAR and 4D radar fusion in various weather conditions," in *Proc. IEEE/CVF CVPR*, 2024, pp. 15162–15172.

[13] J. Luiten, A. Osĕp, P. Dendorfer, P. Torr, A. Geiger, L. Leal-Taixé, and B. Leibe, "HOTA: A higher order metric for evaluating multi-object tracking," *International Journal of Computer Vision*, vol. 129, pp. 548–578, 2021.

[14] M. Awais, M. Naseer, S. Khan, R. M. Anwer, H. Cholakkal, M. Shah, M.-H. Yang, and F. S. Khan, "Foundation models defining a new era in vision: A survey and outlook," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 4, pp. 2245–2264, 2025.

[15] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking," in *Proc. IEEE ICIP*, 2016, pp. 3464–3468.

[16] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in *Proc. IEEE ICIP*, 2017, pp. 3645–3649.

[17] Z. Wang, L. Zheng, Y. Liu, Y. Li, and S. Wang, "Towards real-time multi-object tracking," in *Proc. ECCV*, 2020, pp. 107–122.

[18] Y. Zhang, C. Wang, X. Wang, W. Zeng, and W. Liu, "FairMOT: On the fairness of detection and re-identification in multiple object tracking," *International Journal of Computer Vision*, vol. 129, pp. 3069–3087, 2021.

[19] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE ICCV*, 2017, pp. 2961–2969.

[20] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *Proc. ECCV*, 2014, pp. 740–755.

[21] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "SSD: Single shot multibox detector," in *Proc. ECCV*, 2016, pp. 21–37.

[22] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," arXiv:2004.10934, 2020.

[23] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE ICCV*, 2017, pp. 2980–2988.

[24] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF CVPR*, 2020, pp. 10778–10787.

[25] I. Robinson, P. Robicheaux, M. Popov, D. Ramanan, and N. Peri, "RF-DETR: Neural architecture search for real-time detection transformers," arXiv:2511.09554, 2025.

[26] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "ByteTrack: Multi-object tracking by associating every detection box," in *Proc. ECCV*, 2022, pp. 1–21.

[27] J. Cao, J. Pang, X. Weng, R. Khirodkar, and K. Kitani, "Observation-centric SORT: Rethinking SORT for robust multi-object tracking," in *Proc. IEEE/CVF CVPR*, 2023, pp. 9686–9696.

[28] N. Aharon, R. Orfaig, and B.-Z. Bobrovsky, "BoT-SORT: Robust associations multi-pedestrian tracking," arXiv:2206.14651, 2022.

[29] G. Maggiolino, A. Ahmad, J. Cao, and K. Kitani, "Deep OC-SORT: Multi-pedestrian tracking by adaptive re-identification," in *Proc. IEEE ICIP*, 2023, pp. 3025–3029.

[30] Y. Du, Z. Zhao, Y. Song, Y. Zhao, F. Su, T. Gong, and H. Meng, "StrongSORT: Make DeepSORT great again," *IEEE Trans. Multimedia*, vol. 25, pp. 8725–8737, 2023.

[31] P. Sun, J. Cao, Y. Jiang, R. Zhang, E. Xie, Z. Yuan, C. Wang, and P. Luo, "TransTrack: Multiple object tracking with transformer," arXiv:2012.15460, 2021.

[32] M. Yang, G. Han, B. Yan, W. Zhang, J. Qi, H. Lu, and D. Wang, "Hybrid-SORT: Weak cues matter for online multi-object tracking," in *Proc. AAAI*, vol. 38, no. 7, 2024, pp. 6504–6512.

[33] Y.-H. Wang, J.-W. Hsieh, P.-Y. Chen, M.-C. Chang, H.-H. So, and X. Li, "SMILEtrack: SiMIlarity LEarning for occlusion-aware multiple object tracking," in *Proc. AAAI*, vol. 38, no. 6, 2024, pp. 5740–5748.

[34] Z. Zhu, H. Peng, C. Zhou, L. Jiang, Z. Li, and H. Ling, "C-BIoU: Complete-IoU loss and cluster-NMS for improving multi-object tracking accuracy," *Pattern Recognition*, vol. 146, Art. no. 110006, 2024.

[35] P. Dendorfer, H. Rezatofighi, A. Milan, J. Shi, D. Cremers, I. Reid, S. Roth, K. Schindler, and L. Leal-Taixé, "MOT17: A benchmark for multi-object tracking," arXiv:1803.02706, 2018.

[36] P. Dendorfer, H. Rezatofighi, A. Milan, J. Shi, D. Cremers, I. Reid, S. Roth, K. Schindler, and L. Leal-Taixé, "MOT20: A benchmark for multi-object tracking in crowded scenes," arXiv:2003.09003, 2020.